MOTILAL NEHRU NATIONAL INSTITUTE OF TECHNOLOGY, ALLAHABAD

DATA STRUCTURE LAB (CSN12101)

ASSIGNMENT- 1

1. Write a function sumdiv( ) that finds the sum of divisors of a number. (Divisors of a number are those numbers by which the number is divisible). For example, divisors of 36 are 1, 2, 3, 4, 6, 9,18. A number is called perfect if the sum of divisors of that number is equal to the number. For example, 28 is a perfect number, since 1+2+4+7+14 =28. Write a program to print all the perfect numbers in a given range.
2. You are supposed to entered positive integer through the keyboard; you have to find the binary equivalent of this number using recursion.
3. Write a function that takes a decimal number and base as argument and returns the equivalent number of the given base.
4. Write a program in C to perform the Addition and Subtraction of two matrices.
5. Write a program in C to perform Matrix Multiplication.
6. Write a program in C to display the lower triangular matrix of an array and upper triangular matrix of an array.
7. Write a C program to read and print student details using structure pointer, demonstrate example of structure with pointer.
8. Implement a binary file database of records with structures:
   - Structure Employee { int id; char name[40]; double salary; };
   - Provide functions to append, search by id, update salary, and delete logically (mark as inactive), all using fread/fwrite on a binary file.
9. Write a memory pool allocator using structures and unions:
   - Use an array of union Block { struct { union Block *next; } freeNode; char payload[BLOCK_SIZE]; };
   - Manage a free list; implement pool_malloc/pool_free that hand out fixed size blocks from the pool.
10. You are given an unsorted array with both positive and negative elements. You have to find the smallest positive number missing from the array.
    Example:
    Input: {2, 3, 7, 6, 8, -1, -10, 15}
    Output: 1
11. Imagine you're managing an extensive dataset for a scientific research project. The data contains multiple arrays, each representing a different metric (e.g., temperature readings, pressure readings, etc.). Your task is to implement a program that efficiently copies data from one array to another using pointers, while also handling various edge cases like dynamic memory allocation, ensuring no data loss, and handling arrays of variable lengths. You are also required to handle potential errors in memory allocation and provide detailed feedback to the user, including information about the copying process (such as when data is successfully copied or if an issue arises).
12. Write a C program to return multiple values from function using pointers.

13. Write a program to implement **Linear search** in a single dimensional array.
14. Imagine you are working at a school where the exam scores of students are stored in an array. The students' scores are sorted in ascending order, and you need to help a student find his/her score in the list. To do this, you decide to use the **Binary Search Algorithm**. Binary search is an efficient way to find an item in a sorted list by repeatedly dividing the search interval in half. If the value is less than the item in the middle of the interval, the search continues in the lower half, or if the value is greater, the search continues in the upper half.
15. Suppose you are working in an organization where the employee's salary is stored in an array. However, the salary are not in order, and you need to sort them from the lowest to the highest. As a part of the task, you decide to implement the **Bubble Sort Algorithm**. This algorithm repeatedly steps through the list, compares adjacent elements, and swaps them if they are in the wrong order. The process continues until the array is sorted. You also have to find the number of comparisons and time required to sort the array.